

Programming for Mobile 2

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Flutter

“Flutter is an open-source framework for building beautiful, natively compiled, multi-platform applications from a single codebase.” by Google

<https://flutter.dev/>

<https://docs.flutter.dev/resources/architectural-overview>

Uses DART programming language (OOP)

<https://dart.dev/language>

Its syntax has basically a blend of CPP, Python, Java, and JavaScript.

Flutter

You can use VS Code as IDE

<https://code.visualstudio.com/>

<https://marketplace.visualstudio.com/items?itemName=Dart-Code.flutter>

<https://docs.flutter.dev/install>

Flutter UI

Declarative:

```
TextButton(  
  child:Text(  
    "Text Button",  
    style:TextStyle(  
      color: Colors.green,  
    ),  
  ),  
  onPressed: () {},  
),
```

Visual Drag & Drog to Generate Code: <https://app.flutterviz.com/#/>

Figma Dev also allows to generate flutter code but it is paid

Flutter

UI

<https://docs.flutter.dev/ui>

Layout

<https://docs.flutter.dev/ui/layout>

Animation

<https://docs.flutter.dev/ui/widgets/animation>

Flutter

Touch

<https://docs.flutter.dev/ui/widgets/interaction>

Sensors

<https://fluttergems.dev/sensors/>

Camera

<https://pub.dev/packages/camera>

Video

<https://docs.flutter.dev/cookbook/plugins/play-video>

QT

<https://www.qt.io/>

The Qt framework is available under both open source and commercial licenses

When selecting an open source license for your project, you contribute to free and open source software development by using Qt under any of the following licenses: LGPL version 3, GPL version 2 and GPL version 3.

QT

QML (Qt Meta-object Language) is a user interface markup language
-> a declarative language

<https://doc.qt.io/qt-6/qmlreference.html>

QML has a signal and handler mechanism, where the *signal* is the event and the signal is responded to through a *signal handler*. When a signal is emitted, the corresponding signal handler is invoked. Placing logic such as a script or other operations in the handler allows the component to respond to the event.

<https://doc.qt.io/qt-6/qtqml-syntax-signals.html>

QT

QT Creator IDE

<https://doc.qt.io/qt-6/topics-app-development.html>

QT Widgets Designer

<https://doc.qt.io/qt-6/designer-to-know.html>

QT Design Studio

<https://doc.qt.io/qtdesignstudio/index.html>

QT

Qt Designer

A simple, lightweight tool for designing Qt Widgets-based Uis. Produces .ui files (XML-based), which are compiled into C++ or used with Python (PyQt/PySide)

Qt Design Studio

A professional UI/UX design and prototyping tool focused on Qt Quick / QML-based Uis. Produces .qml files and full QML projects

	Qt Designer	Qt Design Studio
UI Framework	Qt Widgets	Qt Quick / QML
Target Users	Developers	Designers + Developers
Animation Support	Minimal	Full timeline editor
Design Tool Import	✗	✓ (Figma, Sketch, etc.)
Prototyping	✗	✓
Standalone App	Yes (also in Qt Creator)	Yes (separate product)

QT

QT Sensors

<https://doc.qt.io/qt-6/qtsensors-cpp.html>

QT Touch

<https://doc.qt.io/qt-6/qtouchevent.html>

QT Multimedia

<https://doc.qt.io/qt-6/qtmultimedia-index.html>

Unity

<https://unity.com/>

A cross-platform game engine (C#)

UI

<https://docs.unity3d.com/Manual/ui-systems/introduction-ui-toolkit.html>

Unity Sensors

<https://docs.unity3d.com/Packages/com.unity.inputsystem@1.19/manual/Sensors.html>

Unity Touch

<https://docs.unity3d.com/Packages/com.unity.inputsystem@1.19/manual/Touch.html>

Unity

Webcam

<https://docs.unity3d.com/6000.3/Documentation/ScriptReference/WebCamTexture.html>

Video

<https://docs.unity3d.com/6000.4/Documentation/Manual/VideoPlayer.html>